RESEARCH PORTFOLIO

Robotic Mechanisms, Actuation & System Integration

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The Chinese University of Hong Kong
Expected graduation: May 2027

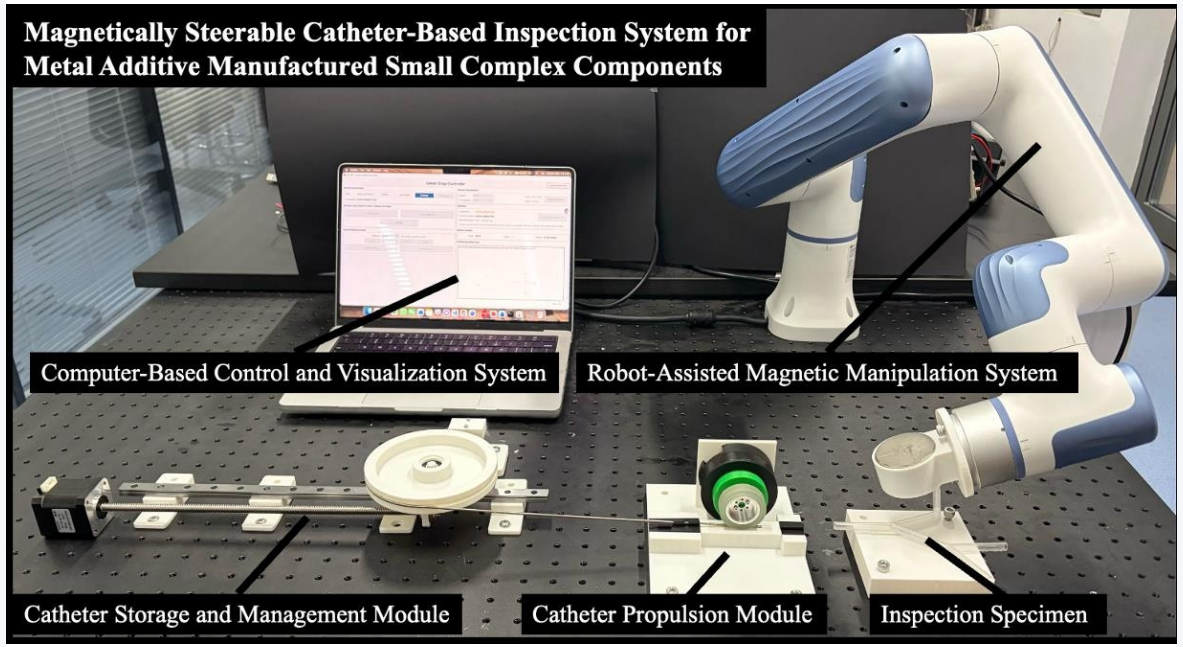
Selected projects in mechanical design, embedded control, magnetic and soft actuation, and experimental validation.

RESEARCH INTERESTS

Cable-driven mechanisms, compliant robotics, and wearable systems.

PHD APPLICATIONS

Fall 2027
Hardware development - actuation - sensing - system integration



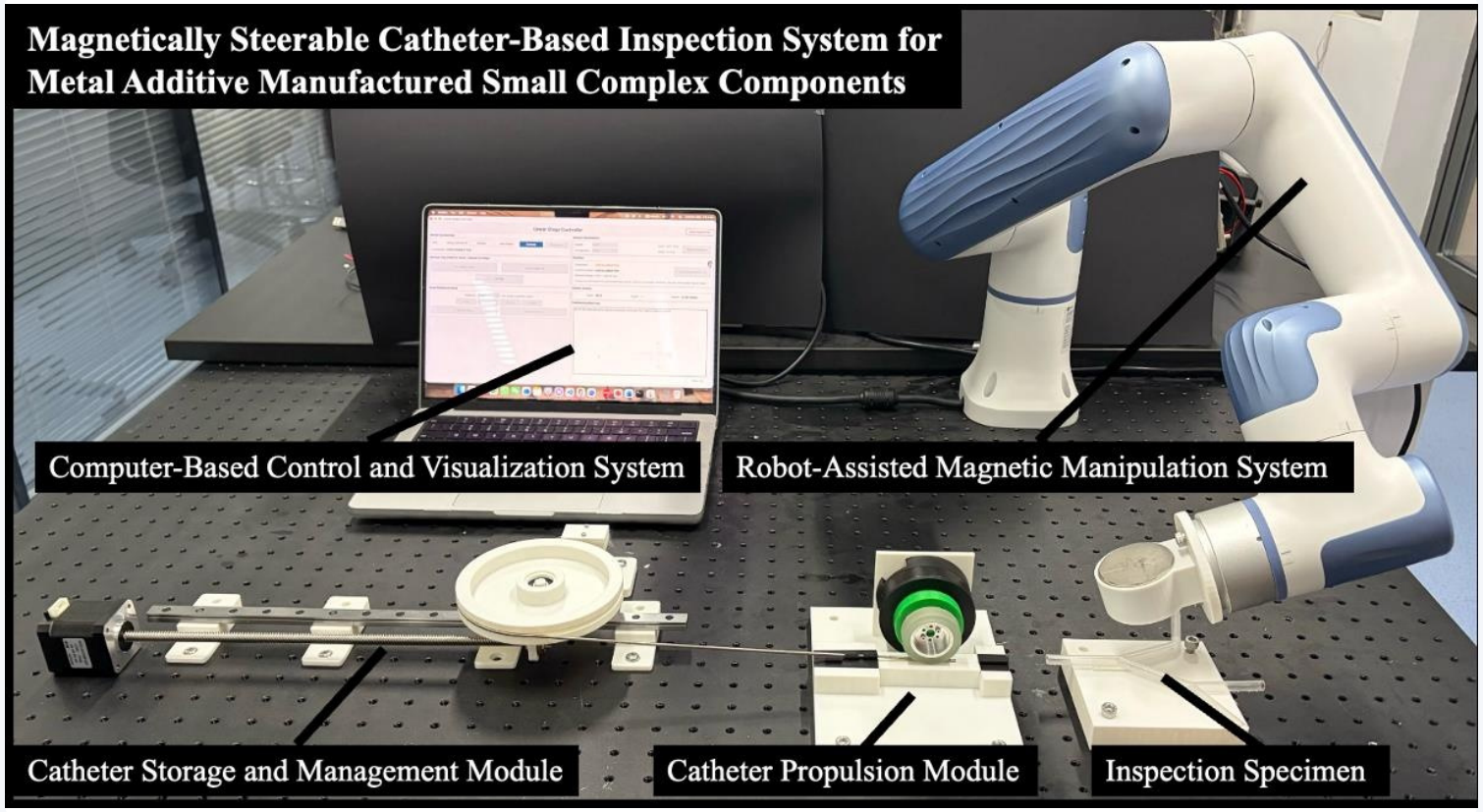
Industrial-inspection setup at XJTU. My contribution: the motorized catheter-management module and its embedded/PC control software.

CUHK | Xi'an Jiaotong University | ETH Zürich

PROJECT 01 | XJTU | JULY-AUGUST 2026

Motorized Catheter Management for Industrial Inspection

Research Assistant | Prof. Qiji Ze's group, Xi'an Jiaotong University



The photograph includes team infrastructure for robot-assisted magnetic manipulation and inspection. My work focused on the catheter-management module and its control software.

MECHANISM DESIGN

Designed and built a translating-pulley catheter accumulator; approximately $\Delta L \approx 2\Delta x$ length compensation. Fusion 360 / SolidWorks assembly with a 120-mm guide pulley, 608 bearings, MGN12H carriage, T8 lead screw, and 3D-printed interfaces.

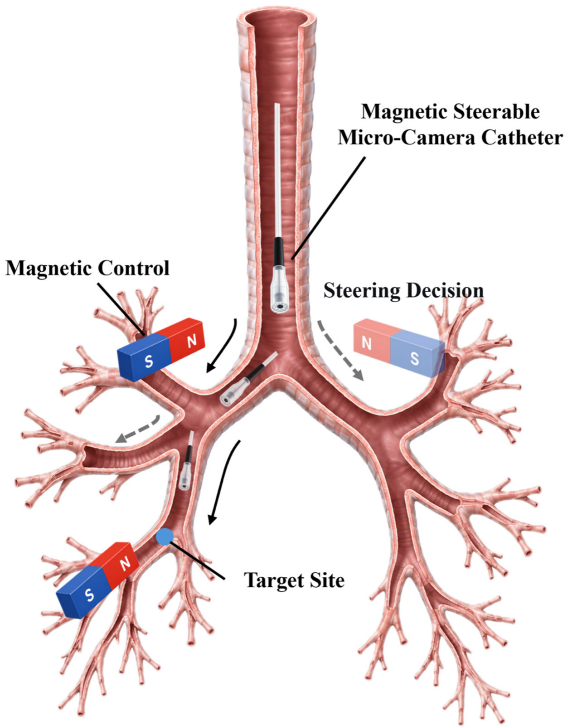
COMPLETE CONTROL CHAIN

Python/Tkinter → USB serial → Arduino UNO → TB6600 → stepper-driven stage. Calibration, jogging, fixed-distance motion, adjustable speed/acceleration, travel limits, and a communication watchdog.

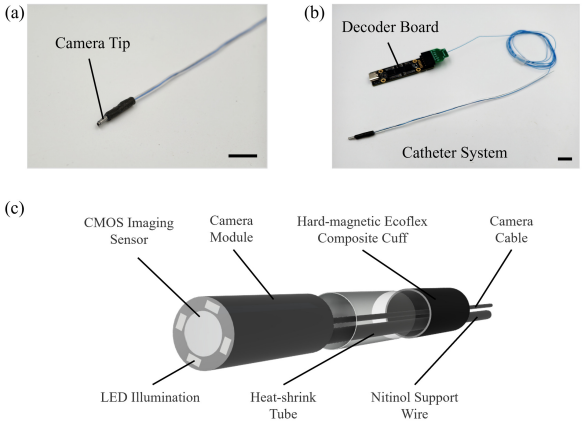
Foundation for further work in robotic mechanisms, motion transmission, and actuation systems.

Magnetically Steerable Micro-Camera Catheter

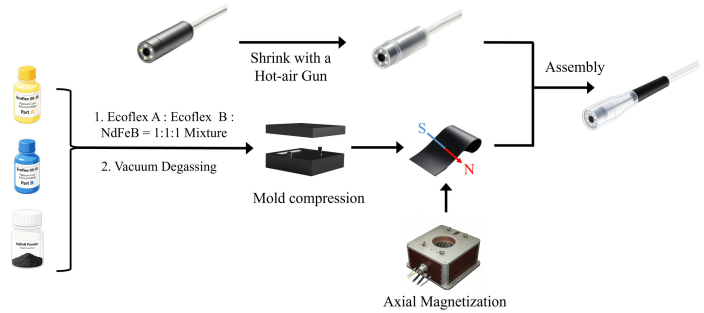
Co-first author | Accepted at ICBIR 2026 | Advisor: Prof. Hongliang Ren



Design concept for bronchial-tree phantom navigation



Prototype structure



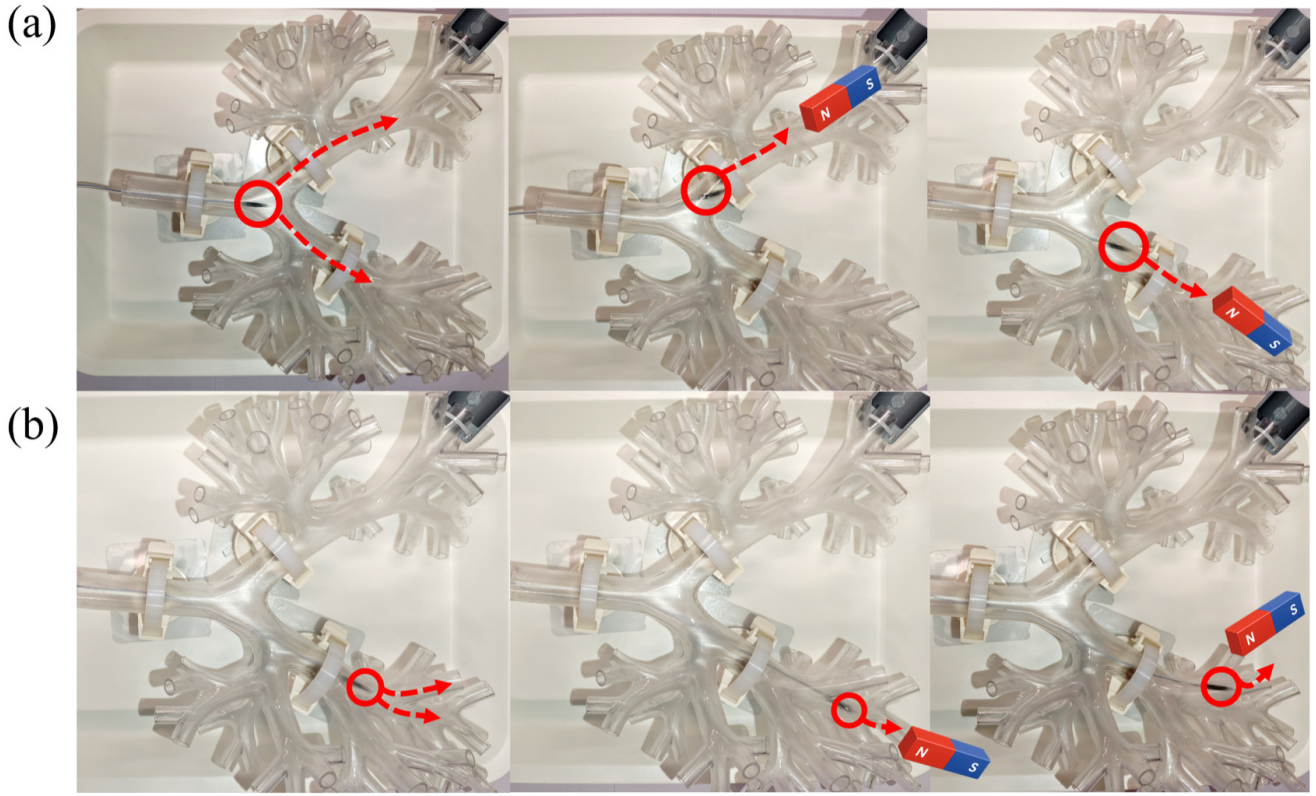
Composite preparation, magnetization, and distal assembly

MY CONTRIBUTION

Designed and fabricated the millimeter-scale catheter; integrated the LED micro-camera, 0.2-mm nitinol support wire, heat-shrink encapsulation, and hard-magnetic Ecoflex cuff. Developed insertion-force and steering-characterization setups and contributed to manuscript and figure preparation.

Magnetic Steering and Forward-Looking Imaging

Bronchial-tree phantom experiments | Manual external permanent-magnet actuation



Magnetically assisted branch redirection. Image sequence reproduced from existing experimental materials.

INSERTION CHARACTERIZATION

Characterized insertion force under dry and saline-wetted conditions.

STEERING AND NAVIGATION

Evaluated multidirectional magnetic steering and forward-looking imaging in a bronchial-tree phantom, including primary and secondary bifurcations.

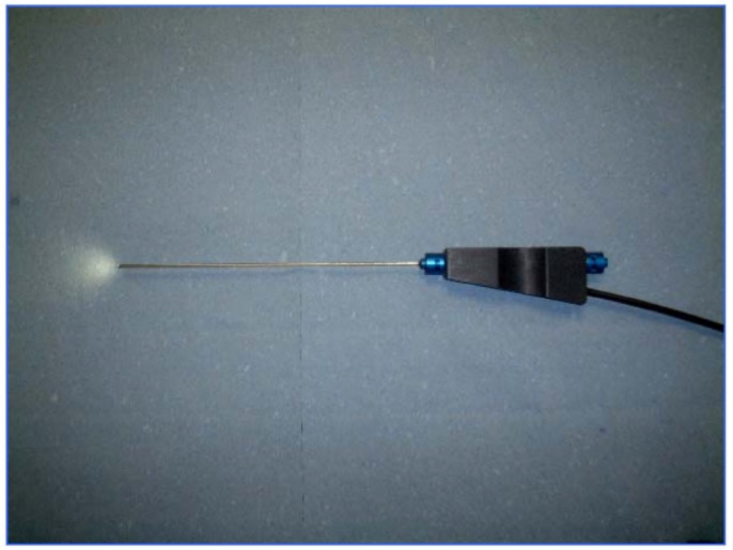
SCOPE OF EVIDENCE

Benchtop phantom evaluation. Further work includes controlled magnetic actuation, visual feedback, and repeated quantitative navigation tests.

PROJECT 03 | ETH ZÜRICH | MARCH-JUNE 2026

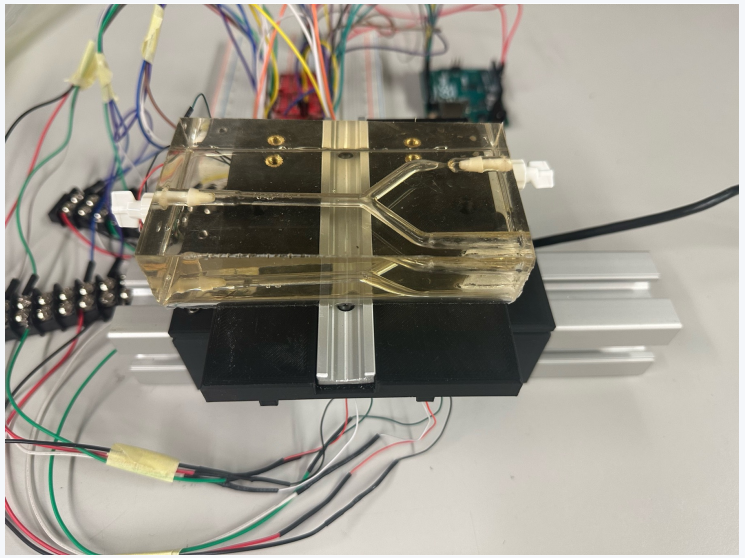
Soft Miniaturized Intraductal Imaging Catheter

Multi-Scale Robotics Lab | Prof. Bradley Nelson | Semester-project grade: 5.5/6.0



FLEXIBLE PROTOTYPES

Designed and fabricated flexible imaging catheter prototypes using 0.5-mm and 0.7-mm nitinol support wires, investigating flexibility, pushability, and insertion resistance. These are support-wire diameters.



CALIBRATED FORCE MEASUREMENT

Built and calibrated a four-load-cell platform with Arduino/Python acquisition. The semester-project report describes 20 manual trials per condition, with 10 selected for analysis; a 1.1-mm rigid reference was also tested.

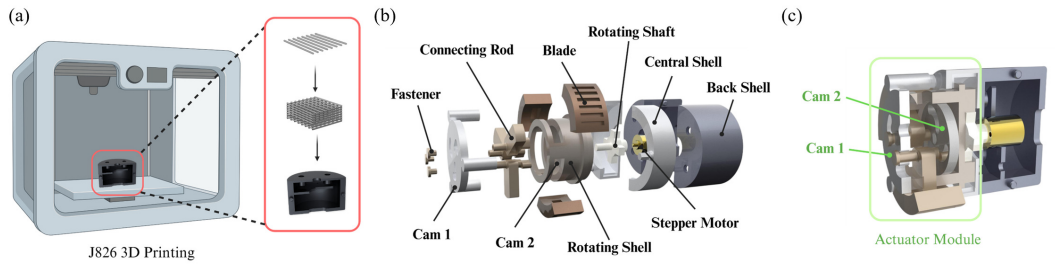


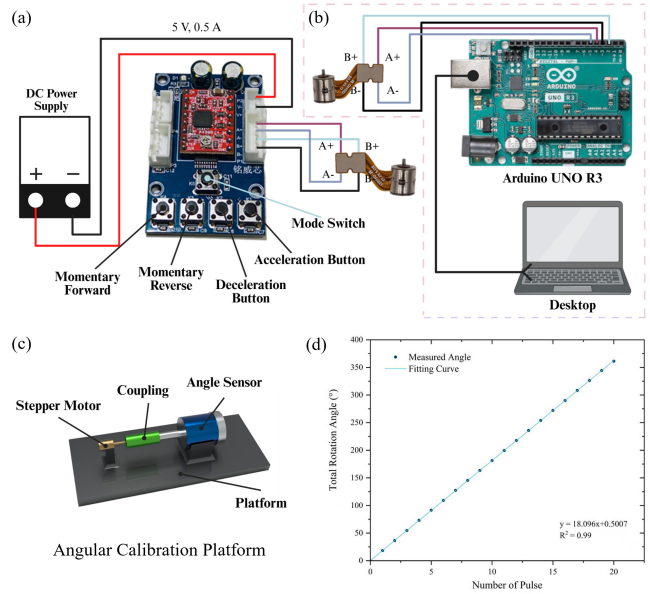
EXPERIMENTAL COMPARISON

Compared support and surface configurations. Fabricated a silicone breast-duct phantom and evaluated imaging with external illumination, including a porcine-blood condition. Results are limited to simplified benchtop models.

Kirigami Capsule for Gastrointestinal Biopsy

Contributing author | Accepted at IEEE ICRA 2026 | Advisor: Prof. Hongliang Ren





SYSTEM INTEGRATION AND EX VIVO VALIDATION

Contributed to iterative assembly, ex vivo porcine experiments, quantitative analysis, and validation writing. Team prototype: 17-mm diameter and 22-mm length.

MOTOR CONTROL AND ANGULAR CALIBRATION

Developed motor-control electronics and Arduino pulse-control logic. Built an angular calibration platform (approximately 18.1 degrees per pulse; $R^2 = 0.99$).

Ruizhou Zhao†, Yichen Chu†, Shuwei Zhao, Wenchao Yue, Raymond Shing-Yan Tang, Hongliang Ren. †Equal contribution.

Author manuscript: arxiv.org/abs/2602.06207. Ex vivo prototype evaluation does not establish clinical safety.

From Hardware Integration to Robotic Assistance

EXISTING EXPERIENCE | PNEUMATIC SOFT ROBOTICS

Designed and fabricated a double-layer spherical-cap silicone actuator and Arduino pneumatic control for benchtop jumping. Developed geometric models and investigated actuator radius and opening angle, deformation, and elastic-energy storage. Ongoing since September 2025.

EXISTING WORK | IMAGING AND DELIVERY CATHETER

Ongoing integration of forward imaging, magnetic steering, and a dedicated delivery channel in a flexible gastrointestinal catheter. Prototype development and experimental evaluation remain in progress.

INTENDED PHD RESEARCH DIRECTIONS

I am interested in extending my experience in mechanism design, actuation, and embedded control toward cable-driven and wearable robotic systems. Potential directions include modular actuation interfaces, lightweight compliant mechanisms, and sensing and control for physical assistance.

These are intended PhD research directions rather than completed projects.

I aim to deepen my expertise in modeling, control, and learning-based methods while building on my hardware and system-integration experience.